Unspoken: Experience and Pages Spec

Nov 12, 2026 · @Janani

At a glance

Unspoken is a $15-a-month mobile app where friend groups, called pods, get an AI planner that finds plans around NYC, polls the group, books with approval, and splits the bill through PayPal. This spec covers the experience and every page, so the backend team knows what to build.

Area	Locked decision
Platform	Standalone mobile app. NYC only for the demo
Pricing	$15 monthly subscription
Pods	A pod is a friend group, 3 to 8 people, created from the phone contact list
Chat	Built-in group chat in every pod. The AI planner lives inside it
Profile	Bio, phone number, interests, and private limits, all editable later
Private limits	Budget and similar limits are set in the profile and never shown to the group
Budget conflicts	If no plan fits someone's budget, the planner asks that person alone, in their private chat, whether a one-time different budget is okay
Voting	Poll votes are visible to the pod
Booking	The planner asks for confirmation before booking and sends reminders until it gets one
Money	When members say yes, one person pays and the others owe them, tracked in PayPal-linked balances

Three things from the earlier backend spec no longer hold: holding everyone's share before charging, hidden votes, and counts-only progress. Section 12 lists what the team should change.

Core rules

Three rules shape every screen: what members pay to use the app, who can see what, and how money moves after a plan is chosen.

Subscription

The app costs $15 a month. Whether each member pays or one member covers the pod is open (section 12). Subscription billing and hangout money are separate: the $15 goes through the phone's app store billing, and PayPal handles only money between friends and venues.

Who sees what

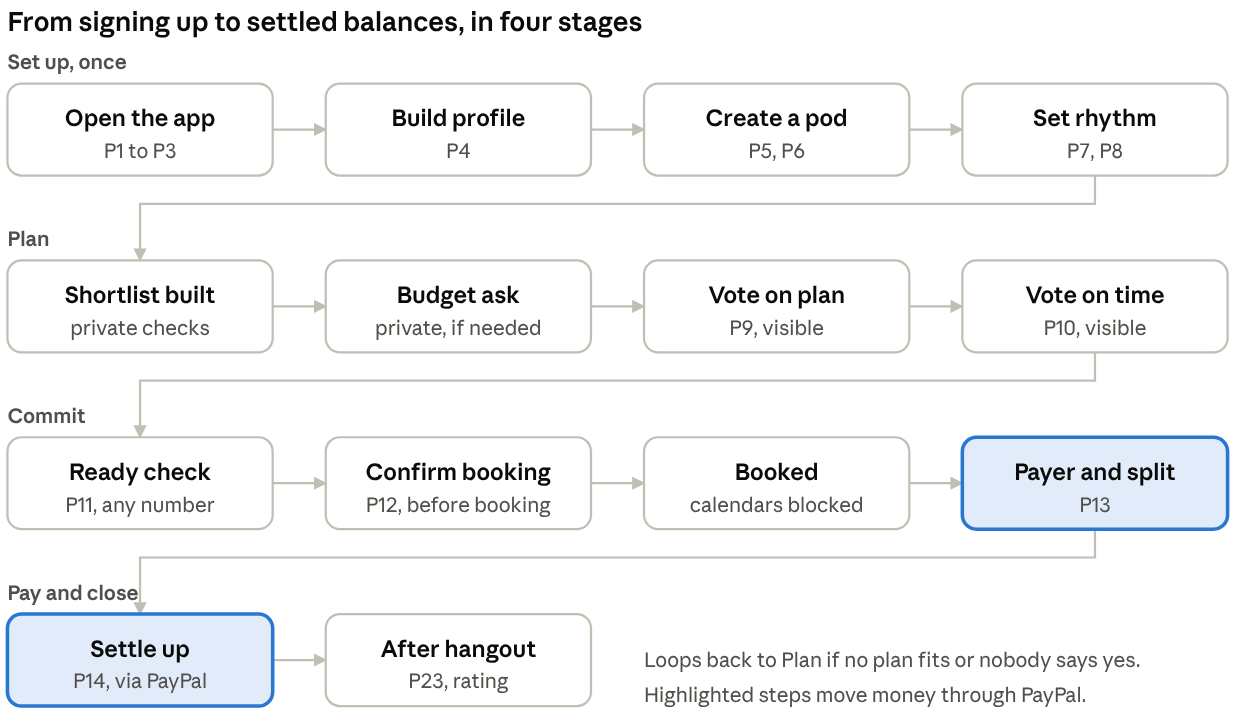
Information	Who sees it	Status
Budget and other private limits	You and the planner. Never a pod, a poll, or a balance	Decided
Budget stretch request	You alone, in your private chat with the planner	Decided
Poll votes	The whole pod, with names	Decided
Calendars	The planner reads only free or busy time to find slots. Members never see each other's calendars	Recommended
Pod chat	Pod members only	Decided
Balances	You see what you owe and what you are owed. Whether the pod sees the full ledger is open	Open

How money moves

1. The pod votes on a plan and a time. Members who want to go say yes. Any number of yeses is enough.
2. The planner asks for confirmation, then books.
3. One person pays the venue from their own PayPal account.
4. The app splits the total across everyone who said yes. Each of them owes the payer their share, shown as a balance.
5. Members settle through PayPal. The app sends reminders until every balance is clear.

Carried over from the backend spec: amounts are integer cents, every PayPal call is idempotent, webhooks are verified, the ledger is append-only, and no AI is in the payment path. Screens show whole dollars for budgets and dollars and cents for money owed.

End-to-end user flow



user flow · 14 steps in four stages

A new user passes through setup once. After that each hangout runs through Plan, Commit, and Pay and close, and the planner loops back to Plan when no option fits or nobody says yes. Page numbers match the page map in the next section.

Page map

The app has 23 screens in five groups. Page numbers below are used throughout this spec.

#	Page	Group	Purpose
P1	Splash	Entry	Animated logo on every open
P2	Sign in	Entry	Verify the phone number
P3	Subscription	Entry	Start the $15 monthly plan
P4	Profile creation	Entry	Bio, interests, private limits, calendar, PayPal, permissions
P5	Home	Home	Floating pods, plus button, hub entry
P6	Create a pod	Home	Pick contacts, name the pod, invite
P7	Pod chat	Pod	Group chat with the planner inside it
P8	Pod settings	Pod	Members, meeting rhythm, group activities
P9	Activity shortlist and vote	Pod	Planner's list of activities, visible votes
P10	Time poll	Pod	Planner's best days and times, visible votes
P11	Ready check	Pod	Members say whether they are in
P12	Booking confirmation	Pod	Planner asks before booking, then reminds
P13	Payer and split	Money	Who pays and each person's share
P14	Settle up	Money	Pay what you owe through PayPal
P15	Money page	Money	All balances across pods
P16	Events search	Hub	Browse NYC events and activities
P17	Event detail	Hub	One event, with suggest-to-pod
P18	Calendar	Hub	Personal calendar and connected calendars
P19	Profile	Hub	Bio, number, interests, private limits
P20	Settings	Hub	Notifications, permissions, subscription, privacy

#	Page	Group	Purpose
P21	Payment methods	Hub	Linked PayPal account
P22	Private planner chat	Hub	One-to-one chat with the planner, including budget asks
P23	After the hangout	Pod	Rating and a shared memory

Pages: entry and onboarding (P1 to P4)

A new user passes through these four pages once. A returning user goes from the splash straight to Home.

P1 Splash

- **On the page:** the app logo animates and turns.
- **Next:** P2 if signed out, P5 if signed in.
- **Backend:** checks the session and preloads the user's pods.
- **Motion:** the logo turn is the first designed animation. Assets to come.

P2 Sign in

- **On the page:** phone number field, code entry, links to terms and privacy.
- **Actions:** request a code, enter it, continue.
- **Backend:** phone verification and a session token. The sign-in method is not yet chosen (section 12).

P3 Subscription

- **On the page:** the $15-a-month plan, three lines on what it includes (the planner, polls and booking, PayPal splitting), a start button, and restore purchase.
- **Actions:** start the plan through the phone's app store billing.
- **Backend:** verifies the purchase receipt and stores plan status. A free trial and who pays are open (section 12).

P4 Profile creation

A stepper of six short screens. Every answer here is editable later from P19.

1. **Basics.** Name, photo, bio, and phone number (prefilled from sign-in).
2. **Interests.** The user picks everything they like doing: activities, cuisines, and the kinds of parties or events they go to. These feed the planner. Backend needs a fixed interest list for this (section 12).
3. **Private limits.** Budget per hangout in whole dollars, dietary needs, dislikes, and longest travel time. A persistent lock badge reads "Only you and your planner see this."
4. **Calendar.** Connect Google Calendar. Other calendars are open (section 12).
5. **PayPal.** Link a PayPal account. Required before the user can pay or be paid in a hangout.
6. **Permissions.** Contacts (to build pods), notifications (polls and reminders), calendar. Each gets a one-line reason. Declining contacts leaves invite links as the only way to build a pod.

Backend: a profile record, interest tags, an encrypted per-user store for private limits that only that user's planner can read (the Constraint Vault from the backend spec), calendar and PayPal account links, and permission status.

Pages: home and pods (P5 and P6)

P5 Home

The home screen is a field of organic bubbles floating at different positions. Each bubble is one pod.

- **On a bubble:** pod name and member count. A quiet state cue shows where the pod's current hangout stands (none, planning, poll open, booking pending, payments open). No cue ever shows a single member's status.
- **Plus button:** opens P6.
- **Hub entry:** a persistent way into the hub: Events search (P16), Calendar (P18), Profile (P19), Settings (P20), Payment methods (P21), Money (P15), and the private planner chat (P22). Its placement is a design choice.
- **Private chat badge:** a dot when the planner has asked the user something privately, such as a budget stretch.
- **Tap a bubble:** opens that pod's chat, P7.
- **Empty state:** no pods yet, so one large bubble invites the user to create the first.
- **Backend:** returns the user's pods with a one-line stage summary and updates it live.

- **Motion:** bubble drift and the pod-created animation are placeholders for later assets.

P6 Create a pod

1. The user taps the plus button. If contacts permission is missing, the app asks for it and explains why.
2. A searchable contact list appears. Contacts already on the app can be selected. Others can be invited with a link (confirm in section 12).
3. The user selects friends. A pod needs 3 to 8 people including the creator, as in the backend spec.
4. The user names the pod and creates it.
5. Invited friends get a notification and accept to join (recommended).
6. The new bubble appears on Home with a birth animation, and the pod chat opens.

- **If contacts permission is declined:** the user can still create a pod and share an invite link.
- **Backend:** matches contacts to accounts by phone number, stores pod and membership records, creates single-use invite links, and enforces the 3 to 8 limit.

Pages: inside a pod (P7 to P12)

Everything the planner does for a pod happens inside the pod chat as cards, so the group sees one conversation.

P7 Pod chat

- **On the page:** pod name and members at the top, the message list, a composer, and a line showing the pod's next hangout.
- **Planner cards in the stream:** the activity shortlist (P9), the time poll (P10), the ready check (P11), the booking confirmation (P12), and the payer and split (P13). The planner appears as its own participant.
- **Actions:** send messages, open a card, tap the header for pod settings (P8).
- **Backend:** real-time messaging with push notifications, message storage, and a way for the planner to post structured cards. Chat extras such as photos, reactions, and read receipts are open (section 12).

P8 Pod settings

- **Members:** names and photos, with leave pod and, for the creator, add or remove.
- **Meeting rhythm:** how often the pod meets. A member proposes a rhythm in the chat, such as weekly, every two weeks, or monthly. When the pod agrees, it becomes the default. Any member can propose a change later the same way. How agreement is reached is open (section 12).
- **Group activities:** specific things the pod wants to do, added by any member. The planner prefers these over general interests.
- **Mute:** silence notifications for this pod.
- **Backend:** stores the rhythm, the next hangout date, and the group activity list. The planner starts planning on the rhythm date.

P9 Activity shortlist and vote

When the rhythm date arrives, the planner posts a shortlist of NYC activities in the chat.

- **On each card:** title, venue or organizer, date and time for events, about how much each person pays, tags, and a vote button.
- **How the list is built:** from members' profile interests and the pod's group activities, then checked against every member's private budget, dietary needs, dislikes, and travel limit. The group never sees why an option is missing.
- **Budget stretch:** if no good option fits one member's budget, the planner sends that member a private message in P22 asking whether a different budget is okay for this time only. The group never sees the ask or the answer.
- **Votes:** visible to the pod, with names.
- **No labels:** no card says "budget-friendly" or "why this one."
- **Backend:** an NYC events and venues catalog (seeded for the demo), a ranker, and a private check per member that returns only works or doesn't work.

P10 Time poll

- **On the card:** the best days and times that fit the pod, found from connected calendars, a vote button on each, and a closing time.
- **Actions:** vote, change a vote before it closes.
- **Votes:** visible to the pod, with names. Members are notified when the poll opens and reminded before it closes.
- **No slot works:** the planner offers to widen the window.

- **Backend:** reads free or busy time from connected calendars, ranks slots, creates the poll, and schedules the closing and reminders. Whether the activity and time polls are one poll or two is open (section 12).

P11 Ready check

- **On the card:** the chosen plan and time, and a yes or no for each member.
- **Rule:** any number of yeses is enough to go ahead. Members who say yes are the ones who share the cost.
- **Why it comes before booking:** the planner needs a head count to book a table or tickets and to split the cost.
- **If nobody says yes:** the plan closes and the planner offers the next-best option.
- **Backend:** stores each member's answer and passes the head count to booking and to the split. This order is recommended and is flagged in section 12.

P12 Booking confirmation

- **On the card:** what will be booked, for how many, when, and the estimated cost, with a confirm button.
- **Rule:** the planner books nothing until it gets confirmation, and sends reminders until it does. Who confirms is open (section 12).
- **Status chip:** waiting for confirmation, booking, booked, or could not book. A failed booking offers the next-best plan.
- **Restaurants:** the planner reserves a table for the chosen time.
- **Parties and events:** the planner may contact the organizer to ask about ticket prices. It asks for the same confirmation before it sends anything.
- **After booking:** the event is blocked on connected calendars. Whose calendars is open (section 12).
- **Backend:** reservation and ticket outreach, with a simulated booking path for seeded NYC venues in the demo, status tracking, reminder jobs, and calendar writes.

Pages: money and after the hangout (P13 to P15, P23)

PayPal is the center of this part of the app. One member pays the venue, and the members who said yes each owe that member their share.

P13 Payer and split

- **On the card:** the total cost, who pays, and each yes-member's share. Nobody who said no appears in the split.
- **Choosing the payer:** open (section 12). The payer must have a linked PayPal account.
- **Split:** equal by default. Shares are rounded so they add up to the total, with any leftover cents going to the first shares in random order, as in the backend spec. A custom split is open.
- **Estimate and actual:** balances start from the estimate. After the event the payer enters the final amount, and every balance updates to match.
- **Actions:** the payer confirms the split. Members see their share and the payer's name.
- **Backend:** creates one balance per member in the ledger the moment the payer confirms, and notifies each member.

P14 Settle up

- **For someone who owes:** who they owe, for which hangout, how much, and a Pay with PayPal button. The button opens PayPal approval and returns to the app.
- **For the payer:** each person who owes them and whether it is owed, pending, or paid.
- **Statuses:** owed, pending, paid, failed.
- **Reminders:** private to each person who still owes. They never go into the pod chat.
- **Backend:** creates each PayPal payment, confirms it from PayPal's webhook, and writes the result to the ledger. Retries must not charge twice. A failed payment returns the balance to owed and tells the member. Which PayPal product moves money to a friend's account is open (section 12).

P15 Money page

- **On the page:** two totals at the top, what you owe and what you are owed, then a list by hangout and pod.
- **Actions:** tap any line to open P14. Filter by pod. See paid history.
- **Backend:** computes every balance from the append-only ledger and never stores a balance directly.

P23 After the hangout

- **Rating:** one tap, loved it, fine, or not for me. It goes only to the member's own planner and tunes their future suggestions, as in the backend spec.

- **Next step:** the planner shows the next hangout date from the pod's rhythm.
- **Idea, not in your flow:** a shared memory page for each hangout, with photos members add, collected into the pod as a growing record of the group.

Pages: the hub (P16 to P22)

The hub is reached from Home. It holds everything personal to the user: events, calendar, profile, settings, payment account, and the private chat with the planner.

P16 Events search

- **On the page:** a search bar, filters (date, price, category, neighborhood), and a list of event and activity cards across NYC.
- **Actions:** search, filter, open an event (P17).
- **Backend:** an NYC events and activities catalog. The demo uses a seeded list. The production data source is not chosen.

P17 Event detail

- **On the page:** title, venue or organizer, date and time, price, description, a map, and tags.
- **Actions:** suggest to a pod, save, add to calendar.
- **Suggest to a pod:** the user picks a pod, and the event is posted in that pod's chat as a suggestion the planner adds to its next shortlist.
- **Backend:** stores saved events and suggestions per pod.

P18 Calendar

- **On the page:** the user's own calendar in week and month views, showing connected calendars and Unspoken hangouts together.
- **Actions:** connect or disconnect a calendar, choose which calendars the planner may read, and open a hangout.
- **Calendars:** Google Calendar is the planned connection. Apple Calendar and Outlook are open (section 12).
- **What the planner reads:** free or busy time only, as recommended in the core rules.
- **Backend:** calendar account links, free or busy reads, and event writes when a plan is booked.

P19 Profile

- **On the page:** photo, name, bio, phone number, interests, and private limits.
- **Interests:** the same picker as profile creation, editable at any time.
- **Private limits:** budget in whole dollars, dietary needs, dislikes, and travel time, behind a lock badge reading "Only you and your planner see this."
- **Backend:** edits apply from the planner's next run. Private limits stay in the encrypted per-user store.

P20 Settings

- **Notifications:** on or off for each type: polls, reminders, private planner asks, and money.
- **Permissions:** current status of contacts, calendar, and notifications, with a shortcut to the phone's settings.
- **Subscription:** plan status and a link to manage it in the app store.
- **Privacy:** a plain explanation of what the planner can see and what the pod can see.
- **Account:** log out and delete account.

P21 Payment methods

- **On the page:** the linked PayPal account and its status, with link and unlink.
- **Bank details:** the app never stores bank or card numbers. Banks and cards are managed inside PayPal.
- **Backend:** stores the PayPal account link and nothing else about the user's money sources.

P22 Private planner chat

A one-to-one chat between the user and their planner. The pod never sees it.

- **Budget stretch asks:** a card that says what the option costs and asks whether a different budget is okay for this time only. The user taps yes or no. If there is no answer by a deadline, the option is dropped without telling the pod (recommended).
- **Other private asks:** reminders about things waiting on the user, such as a confirmation or a payment owed.
- **Free text:** the user can tell the planner new limits in their own words. The planner shows what it understood as editable chips and saves nothing until the user confirms.

- **Backend:** a private message store per user, stretch ask records that apply to one hangout only, and no write path from this chat into any pod.

Permissions, notifications, and edge cases

Rows marked (recommended) are proposals for you to confirm, not decisions.

Permissions

Permission	Used for	Asked at	If declined
Contacts	Building pods	P4 step 6, or the first tap of the plus button	Pods are built with invite links only
Calendar	Finding times and blocking booked events	P4 step 4	The planner asks the member for times in the private chat (recommended)
Notifications	Polls, reminders, private asks, money	P4 step 6	Reminders show only inside the app
PayPal link	Paying and being paid	P4 step 5	The member can chat and vote but cannot say yes in a ready check until linked (recommended)

Notifications

Event	Goes to	Note
Pod invite	The invited friend	Accept to join
Shortlist posted	The pod	
Time poll opened, closing soon	The pod, then members who have not voted	Reminders are private, never a call-out in the chat
Budget stretch ask	That member only	Lives in P22
Ready check	The pod	

Event	Goes to	Note
Booking confirmation needed	Whoever confirms	Repeats until confirmed
Booked	The pod	Also written to calendars
Balance created	The payer and each member who owes	
Payment received or failed	The payer and the member who paid	
Balance reminder	Each member who still owes	Private, never in the pod chat

Edge cases

Situation	What the app does
No pods yet	Home shows one large bubble inviting the first pod
No plan fits everyone	The planner offers to widen the date window or area. It never says whose limit blocked a plan
Nobody votes by the close	The planner extends once, then closes the poll (recommended)
Tied vote	Lower cost wins, then the earlier start, as in the backend spec (recommended)
Pod drops below 3 members	Open: the 3-member minimum came from the old hidden-vote design (section 12)
Booking fails	The planner says so and offers the next-best plan
Payer drops out after booking	Open: the pod picks a new payer (section 12)
A member says yes, then leaves	Their balance stays until settled
A PayPal payment fails	The balance returns to owed and the member is told
Subscription ends	Pods and history stay readable. The planner stops planning (recommended)

Situation	What the app does
No connection	Chat and cards load from the last sync. Actions that move money wait for a connection

Motion and visual placeholders

The backend does not depend on any of this. Inspiration and assets come after the backend is built, so engineers should leave a place for each moment below and no more. PayPal blue is used only for line drawings, here and there, and never for fills or buttons.

Page	Moment	Source
P1	The logo animates and turns	Your flow
P5	Pods float as organic bubbles in different parts of the screen	Your flow
P5	Bubbles drift toward each other when every member is free (recommended)	Addition
P5 to P7	A tapped bubble opens into its chat	Your flow
P6	A new pod bubble is born	Addition
P5, P8	The pod's meeting rhythm shows as a slow pulse in its bubble	Addition
P9, P10	Cards arrive in the chat and votes settle onto the winner	Addition
P12	A confirmed booking pops or settles the bubble	Addition
P13 to P15	A blue line drawing threads from the payer to each member and fades as balances clear	Addition
Empty states	Blue line drawings for the first pod, no events, and no balances	Addition

More additions for later design

Addition	Where it would appear	What the backend needs
Spontaneous mode: the planner suggests a last-minute plan when calendars open up	P7	Calendar watching and a trigger outside the rhythm
Travel time and weather on activity cards	P9, P17	A travel time source and a weather source
Swipe browsing of events	P16	None beyond the events catalog
Apple Calendar and Outlook connections	P18	Two more calendar integrations

Open decisions and handoff notes

The first table lists decisions only you can make. The second tells the backend team what in the earlier spec no longer applies.

Decisions for you

#	Decision	Recommendation
1	Who pays the $15: each member, or one member for the pod	None yet
2	Free trial before the $15 starts	None yet
3	Sign-in method	Phone number and a text code, since the profile holds a number
4	How the payer pays the venue: through PayPal in the app, or at the venue by any method with the app only tracking what friends owe	Track what friends owe, and let the payer pay the venue however the venue takes payment
5	How the payer is chosen	A volunteer, with the planner suggesting a rotation
6	Who confirms a booking	The payer, since they front the money

#	Decision	Recommendation
7	Order of ready check and booking	Ready check first, because booking needs a head count. The alternative is that a plan vote also counts as the yes
8	One poll or two for activity and time	Two cards in sequence: activity, then time
9	Whose calendars the booked event is blocked on	Only members who said yes
10	How a meeting rhythm is agreed	Every member accepts the proposal
11	Minimum pod size	Keep 3 to 8 for now. The 3-member minimum was set for hidden votes, which no longer apply
12	Contacts who are not on the app	Invite them with a link
13	Chat extras: photos, reactions, read receipts	Photos and reactions yes, read receipts no
14	Who sees which balances	Only the people involved in each balance
15	Whether members can see each other's interest lists	None yet
16	The fixed interest list used in profiles	Design it with the NYC demo events in mind
17	Only one member says yes	The planner checks before going ahead
18	Split type	Equal split only for the first version
19	If the payer drops out after booking	The pod picks a new payer
20	Calendars beyond Google	Apple and Outlook after the demo

What changes from the earlier backend spec

Earlier backend spec	This product	What the backend team does
Each member's share is held, then all are captured together, or all are voided	One member pays and the others owe them. There are no holds	Remove authorize, capture, and void flows. Build balances in the ledger instead
Hangout states COMMITTING, SETTLING, VOIDING, NOT_HAPPENING, SETTLEMENT_FAILED	A new state list is needed	Proposed states: PLANNING, STRETCH_ASKS, ACTIVITY_VOTE, TIME_VOTE, READY_CHECK, AWAITING_CONFIRMATION, BOOKING, BOOKED, BALANCES_OPEN, SETTLED, NO_PLAN, BOOKING_FAILED
Counts-only progress and hidden votes	Votes and yeses are visible with names	Drop the count-only display. Keep the private budget checks, which still return only works or doesn't work
The venue is paid by each friend, or by a platform payout	The payer pays the venue and friends repay the payer	Drop the venue payee and Payouts fallback unless decision 4 brings them back
Coordinator Agent and Personal Agent	Users see one planner	The pod chat is the Coordinator. The private chat is each user's Personal Agent
PayPal Orders, Payouts, and the authorize webhooks	Friends pay a friend's account	Confirm which PayPal product can pay a friend's personal account, and rewrite the go/no-go test for a payment between two sandbox accounts with a webhook and a repeat-safe retry
No subscription	$15 monthly plan	Add app store billing and a plan check on the planner
Group UI and private agent chat	A full chat app	Add real-time messaging, push, contact matching, calendar connections, and an events catalog